Requirements Gathering

E-commerce Supply Chain | Data Engineering Project

Project Status: In Progress | Expected Completion: 4 Weeks

1. Purpose and Project Status

This document defines the preliminary business, functional, data, and non-functional requirements for the E-commerce Supply Chain Data Engineering project. The project is currently in progress, and the team expects to complete it within four weeks. The requirements below are intended to guide the remaining implementation and validation work and should be reviewed against the final agreed scope.

2. Problem Statement

E-commerce supply-chain data may contain inconsistent formats, missing values, duplicates, or fields that need transformation before they can be used reliably. The project aims to create a repeatable data-processing workflow that prepares the selected dataset, stores structured outputs, validates the results, and makes the processed data available for analysis and reporting.

3. Stakeholders

Stakeholder	Needs
Project team / data engineers	Clear requirements, data definitions, repeatable processing, and testable outputs.
DEPI evaluator / supervisor	Well-defined scope, justified methodology, documented testing, and transparent limitations.
Business/operations representative	Consistent data that can support relevant e-commerce supply-chain questions.
Report consumer / analyst	Understandable metrics and reports based on documented definitions.

4. Business Objectives

- Prepare the selected dataset for the agreed analytical use cases.
- Build a repeatable pipeline using Python and Pandas.
- Store and query structured data using PostgreSQL and SQL.
- Orchestrate agreed workflow tasks using Apache Airflow.
- Validate processing outputs and make supported insights available through Power BI.
- Complete the project and documentation within the four-week target.

5. Functional Requirements

ID	Requirement	Priority	Acceptance criteria
FR-01	Document the dataset source, schema, data types, and relevant field meanings.	Must	A dataset description/data dictionary is included.

FR-02	Access the dataset using the agreed input method.	Must	The documented input step can be reproduced by a team member.
FR-03	Clean and transform the data using Python/Pandas according to documented rules.	Must	Transformation logic is documented and output schema matches the agreed design.
FR-04	Load or persist processed structured data in PostgreSQL.	Must	Expected tables/records can be queried using SQL.
FR-05	Use Apache Airflow to define and orchestrate the agreed pipeline tasks.	Must	The DAG is visible and a representative workflow run can be demonstrated.
FR-06	Apply data-quality validation to relevant fields and outputs.	Must	Checks and their outcomes are recorded; rules match the dataset.
FR-07	Use SQL to support required queries, aggregations, and validation.	Must	Queries return expected results on the agreed test cases.
FR-08	Create Power BI reports/dashboards for the approved metrics.	Must	Visuals use documented metric definitions and the processed data source.
FR-09	Test key Python processing functions and rules where feasible, using Pytest.	Should	Selected tests run and results are documented.
FR-10	Support scheduled/repeatable execution through Airflow.	Must	The workflow can be triggered according to the agreed demonstration procedure.

6. Data Requirements

- Record the dataset source and any usage restrictions.
- Identify record grain, field types, key columns, and meanings before transformations.
- Profile missing values, duplicates, inconsistent formats, and invalid values where relevant.
- Document cleaning decisions and any assumptions made.
- Define metrics in terms of available fields and explicit formulas.
- Do not infer stock levels, inventory optimization, delivery prediction, or other capabilities unless the required data and validation support them.

7. Non-Functional Requirements

ID	Requirement	Verification
NFR-01	Reproducibility: the documented pipeline can be run again by the team.	A team member follows the run instructions and checks outputs.
NFR-02	Data integrity: transformations do not cause unexplained record loss or duplication.	Compare row counts and key uniqueness where applicable.

NFR-03	Maintainability: code, SQL, DAGs, and assumptions are understandable.	Peer review of repository structure and documentation.
NFR-04	Reliability: pipeline failures and unexpected outputs can be identified.	Review Airflow run status/logs and validation results.
NFR-05	Usability: report labels and metrics are understandable.	Peer review of Power BI visuals and metric definitions.
NFR-06	Time: project work is completed within four weeks.	Weekly milestone and blocker review.
NFR-07	Security/privacy: sensitive fields, if any, are handled appropriately.	Review dataset fields and document access/storage precautions.

8. Use Cases

Use case	Actor	Main flow	Expected outcome
UC-01: Inspect data	Data engineer	Review source, schema, types, missingness, duplicates, and field meaning.	Dataset profile and data dictionary.
UC-02: Process data	Data engineer	Run Python/Pandas cleaning and transformation steps.	Processed dataset with agreed schema.
UC-03: Store and query data	Data engineer / analyst	Load processed data into PostgreSQL and run SQL queries.	Accessible structured data and query results.
UC-04: Orchestrate pipeline	Data engineer	Trigger/run the Airflow DAG and inspect task outcomes.	Documented workflow run and task status.
UC-05: Validate output	Tester / team member	Run quality checks and selected tests; investigate issues.	Validation evidence and issue log.
UC-06: Review analytics	Analyst / evaluator	Inspect Power BI visuals and metric definitions.	Understandable, traceable reporting outputs.

9. Constraints and Dependencies

- Four-week target for completing remaining project work.
- Dataset fields determine which supply-chain questions can be answered.
- Airflow setup and integration must be tested early enough to leave time for debugging.
- The final output must match the team's implemented scope and available computing environment.
- Team availability and component integration can affect the schedule.

10. Open Questions for Final Review

- Confirm the exact dataset name, source, and schema.
- Confirm the PostgreSQL table design and how data will be loaded.
- Confirm the Airflow DAG tasks, dependencies, and schedule used for the demonstration.
- List the exact Power BI metrics and visuals included.
- Record actual test results, pipeline run evidence, and measured outcomes.

11. Approval

The team should review and approve these requirements, update the status of each item, and mark any optional or infeasible feature as out of scope before final submission. The document describes the planned implementation; completion claims should be made only after the relevant acceptance criteria have been tested.

Project Team

- Menna Hamdy
- Omar Salem
- Ibrahim Elglad
- Belal Yasser
- Ahmed Sami
- Abdalla Elhelaly